

Lab. 2

Linear Op-Amp Circuits

Team No. _____

Names:

_____ I.D. _____
_____ I.D. _____

Reference:

Floyd Thomas L., Buchla David M.: *Experiments in Analog Fundamentals, A System Approach*. Pearson Education, 2013.

Reading:

Floyd and Buchla, *Fundamentals: A Systems Approach*, Sections 6-4 through 6-6. Pearson Education, 2013.

Objectives:

After performing this experiment, you will be able to:

1. Construct and test inverting and noninverting amplifiers using op-amps.
2. Specify components for inverting and noninverting amplifiers using op-amps.

Materials Needed:

- Resistors: two 1 k Ω , one 10 k Ω , one 470 k Ω , one 1.0 M Ω
- Two 1.0 μ F capacitors
- One 741C op-amp
- *For Further Investigation:*
 - One 1.0 k Ω potentiometer, one 100 k Ω resistor, assorted resistors to test.

Summary of Theory

One of the most important ideas in electronics incorporates the idea of *feedback*, where a portion of the output is returned to the input. If the return signal tends to decrease the input amplitude, it is called *negative feedback*. Negative feedback produces a number of desirable qualities in an amplifier, increasing its stability and its frequency response. It also allows the gain to be controlled independently of the device parameters, temperature, or other variables.

Operational amplifiers are almost always used with external, negative feedback. The feedback circuit determines the specific characteristics of the amplifier. By itself, an op-amp has an extremely high voltage gain called the *open-loop* gain, A_{ol} . When negative feedback is added, the overall gain of the amplifier is determined by the feedback circuit. This gain, including the feedback circuit, is called the *closed-loop* gain, A_{cl} .

Figure 20-1 illustrates a noninverting amplifier with negative feedback. The input is applied to the noninverting terminal and a fraction (B) of the output is returned to the inverting input by the voltage divider. The very high gain of the amplifier forces the two inputs to be very nearly the same voltage; therefore, the input voltage is across R_i , and the output voltage is across $R_i + R_f$. The closed-loop gain of this noninverting amplifier is given as $A_{cl(NI)}$. The closed-loop gain is the reciprocal of the feedback fraction, as shown in Figure 20-1.

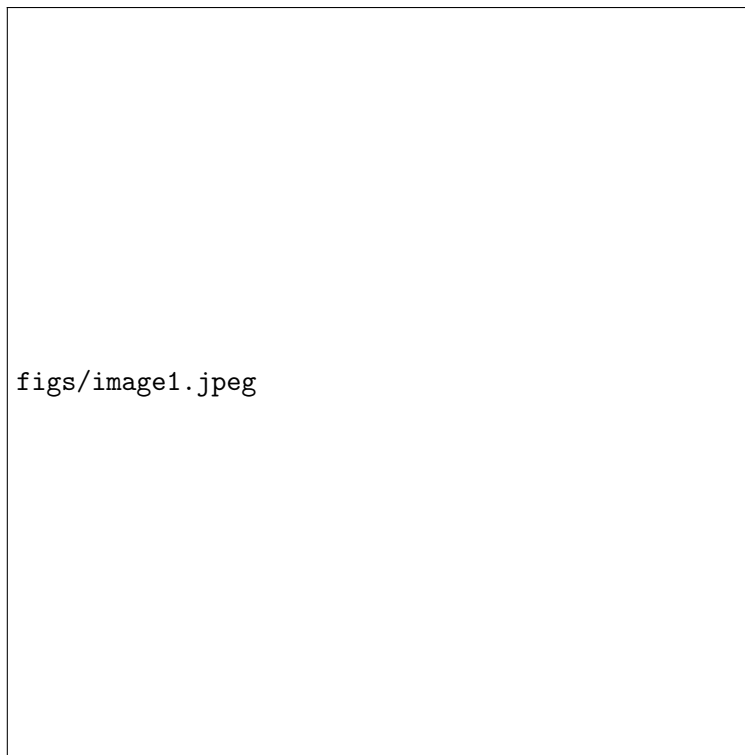


Figure 20-1: Noninverting amplifier

$$\text{feedback fraction} = B = \frac{R_i}{R_i + R_f}$$

$$A_{cl(NI)} = \frac{1}{B} = \frac{R_i + R_f}{R_i} = 1 + \frac{R_f}{R_i}$$

An inverting amplifier is shown in Figure 20-2. In this amplifier, the output voltage is the opposite phase to the input voltage. The noninverting input is grounded, and the input signal is applied through a resistor to the inverting terminal. A feedback resistor is connected between the output and the inverting input. This amplifier can be analyzed by assuming the input current is the same as the current in the feedback resistor, $I_{in} = I_f$, and that the open-loop gain of the amplifier is very high. From these assumptions, the closed-loop gain can be calculated quite accurately as the ratio of the feedback resistor to the input resistor. The basic equations for the inverting amplifier are shown with Figure 20-2. The minus sign indicates inversion between the input and output. $(-)$ refers to the inverting input on the op-amp.

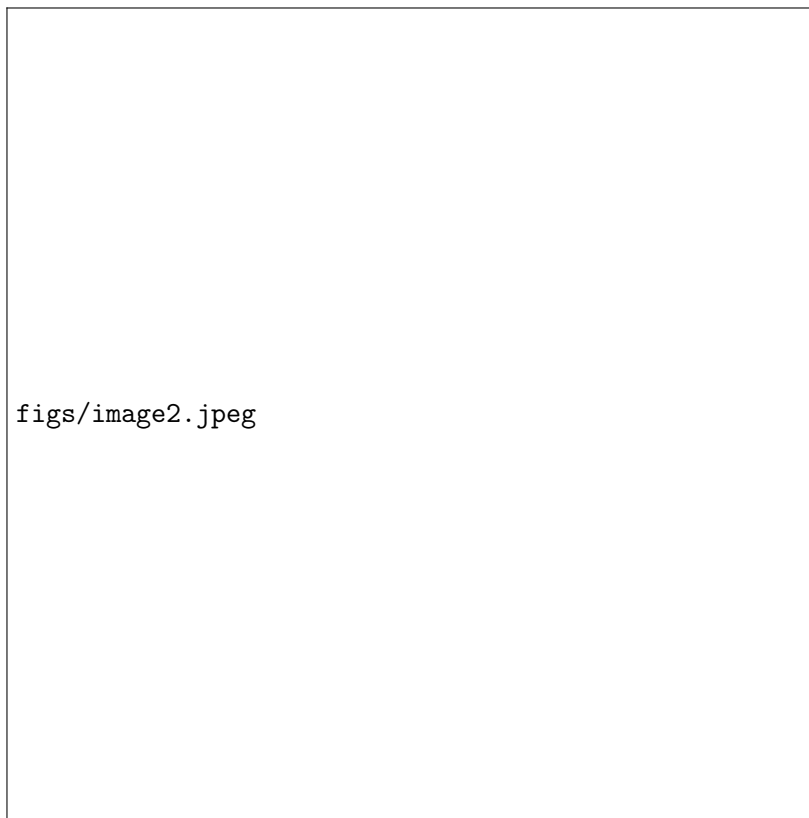


Figure 20-2: Inverting amplifier

$$I_{in} = I_f$$

A_{ol} is very large, therefore

$$V(-) = -\frac{V_{out}}{A_{ol}} \cong 0$$

$$\frac{V_{in}}{R_i} = -\frac{V_{out}}{R_f}$$

$$A_{cl(I)} = \frac{V_{out}}{V_{in}} = -\frac{R_f}{R_i}$$

Procedure

1. The circuit to be tested in this step is the noninverting amplifier illustrated in Figure 20-3.
 - a. Measure a $10\text{ k}\Omega$ resistor for R_f and $1.0\text{ k}\Omega$ resistor for R_i . Record the measured value of resistance in Table 20-1.
 - b. Using the measured resistances, compute the closed-loop gain of the noninverting amplifier. The closed-loop gain equation is given next to Figure 20-1.
 - c. Calculate V_{out} using the computed closed-loop gain.
 - d. Connect the circuit shown in Figure 20-3. Set the signal generator for a 500 mV_{pp} sinusoidal wave at 1.0 kHz . The generator should have no dc offset.
 - e. Measure the output voltage, V_{out} . Record the measured value.
 - f. Measure the feedback voltage at pin 2. Record the measured value.
 - g. Place a $1.0\text{ M}\Omega$ test resistor in series with the generator. Measure the input resistance of the circuit, R_{in} , based on the voltage drop across the test resistor. Use the voltage divider rule to indirectly find the input resistance.

Table 20-1

R_f Measured	R_i Measured	V_{in} Measured	$A_{cl(NI)}$ Computed	V_{out} Computed (pin 6)	V_{out} Measured (pin 6)	$V(-)$ Measured (pin 2)	R_{in} Measured
		500 mV_{pp}					

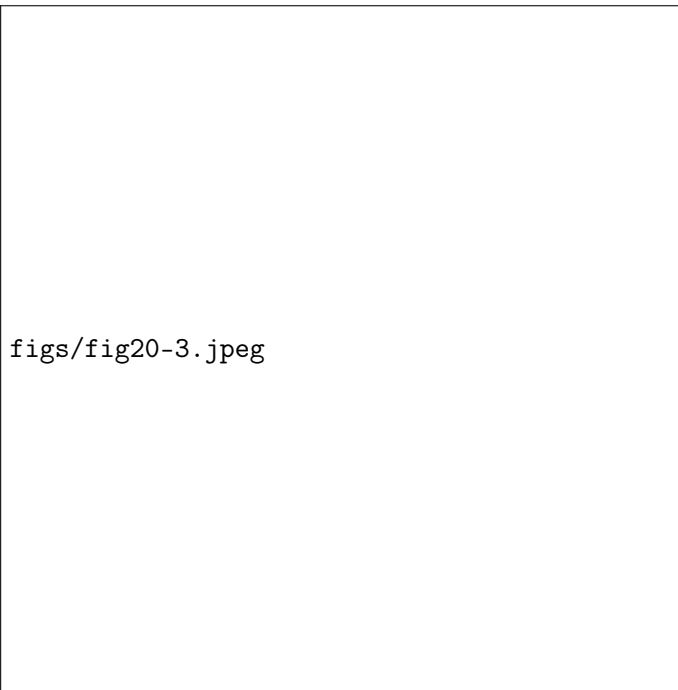


Figure 20-3: Noninverting amplifier test circuit (741C, $R_f = 10\text{ k}\Omega$, $R_i = 1.0\text{ k}\Omega$)

2. In this step you will test an inverting amplifier. All data are to be recorded in Table 20-2. The circuit is illustrated in Figure 20-4. The closed-loop gain is:

$$A_{cl(I)} = -\frac{R_f}{R_i}$$

- a. Use the same resistors for R_f and R_i as in step 1. Record the measured value of resistance in Table 20-2.
- b. Using the measured resistance, compute and record the closed-loop gain of the inverting amplifier.
- c. Calculate the V_{out} using the computed closed-loop gain.
- d. Connect the circuit shown in Figure 20-4. Set the signal generator for the 500 mV_{pp} sine wave at 1 kHz. The generator should have no dc offset.
- e. Measure and record the output voltage, V_{out} .
- f. Measure and record the output voltage at pin 2. This point is called a *virtual ground* because of the effect of negative feedback.
- g. Place a 1.0 kΩ test resistor in series with the generator and R_i . Measure the input resistance of the circuit, R_{in} , based on the voltage drop across the test resistor. In this case, you should observe that the drop across the test resistor is about the same as the drop across R_i .

Table 20-2

R_f Measured	R_i Measured	V_{in} Measured	$A_{cl(NI)}$ Computed	V_{out} Computed (pin 6)	V_{out} Measured (pin 6)	$V(-)$ Measured (pin 2)	R_{in} Measured
		500 mV _{pp}					



figs/fig20-4.jpeg

Figure 20-4: Inverting amplifier test circuit (741C, $R_f = 10\text{ k}\Omega$, $R_i = 1.0\text{ k}\Omega$)

3. In this step you will specify the components for an inverting amplifier using a 741C Op-Amp. The amplifier is required to have an input resistance of $10\text{ k}\Omega$ and a closed-loop gain of -47 . The input test signal is a 1 kHz , 100 mV_{pp} sinusoidal wave signal with no dc component (offset). Check that there is no dc component using an oscilloscope. If necessary, you may need to put a voltage divider on the input to attenuate the signal to the 100 mV level.

Draw the amplifier. Then build and test your circuit.

Note: You need to be careful that the generator does not have a dc offset: remember this is a dc amplifier!

Find the maximum voltage the input signal can have before clipping occurs. Try increasing the frequency and note the frequency at which the output is distorted. Does the upper frequency response depend on the amplitude of the waveform? Summarize your results in the space provided.

Conclusion

Evaluation and Review Questions

1. Express the gain of the amplifiers tested in steps 1 and 2 in dB.

2. It was correct to talk about a virtual ground for the inverting amplifier. Why isn't it correct to refer to a virtual ground for the noninverting amplifier?

3. If $R_f = R_i = 10\text{ k}\Omega$, what gain would you expect for:
 - a. A noninverting amplifier?
 - b. An inverting amplifier?

4. For the noninverting amplifier in Figure 20-3:
 - a. If $R_f = 0$ and R_i is infinite, what is the gain?

 - b. What is this type of amplifier called?

5. What output would you expect in the inverting amplifier of Figure 20-4 if R_f were open?

For Further Investigation

An interesting application of an inverting amplifier is to use it as the basis of an ohmmeter for high value resistors. The circuit is shown in Figure 20-5. The unknown resistor, labeled R_x , is placed between the terminals. The output voltage is proportional to the unknown resistance.

Calibrate the meter by placing a known $10\text{ k}\Omega$ resistor in place of R_x and adjusting the potentiometer for exactly 100 mV output. The output then represents $10\text{ mV}/1000\Omega$. By reading the output voltage, and moving the decimal point, you can directly read resistors from several thousand ohms to over $1\text{ M}\Omega$.

Construct the circuit and test it using different resistors. Calibrate output voltage against resistance and compare with theory. Find the percent error for a $1\text{ M}\Omega$ resistor using a lab meter as a standard.

Summarize your results in a short report.

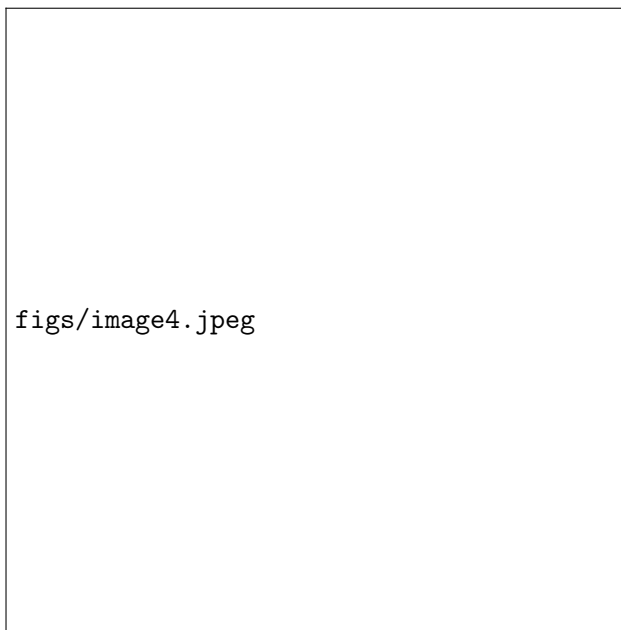


Figure 20-5: Ohmmeter circuit based on the inverting amplifier